



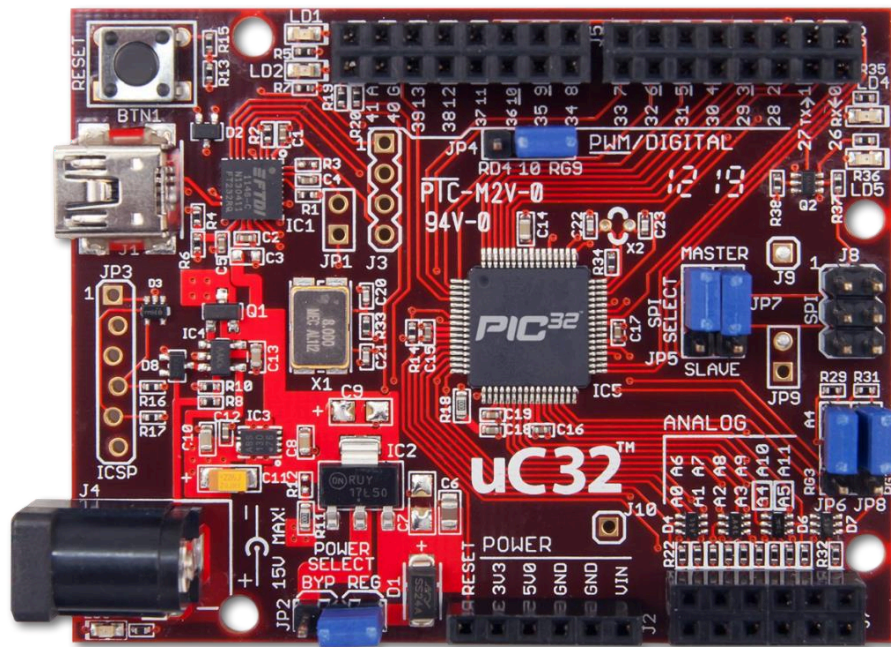
HITEC UNIVERSITY, TAXILA

DEPARTMENT OF COMPUTER ENGINEERING

**Lab # 10: Getting Started with uC32 ChipKIT Board
using Arduino IDE**

Embedded Systems

Getting Started with uC32 ChipKIT Board using Arduino IDE



Dated:

Week 10

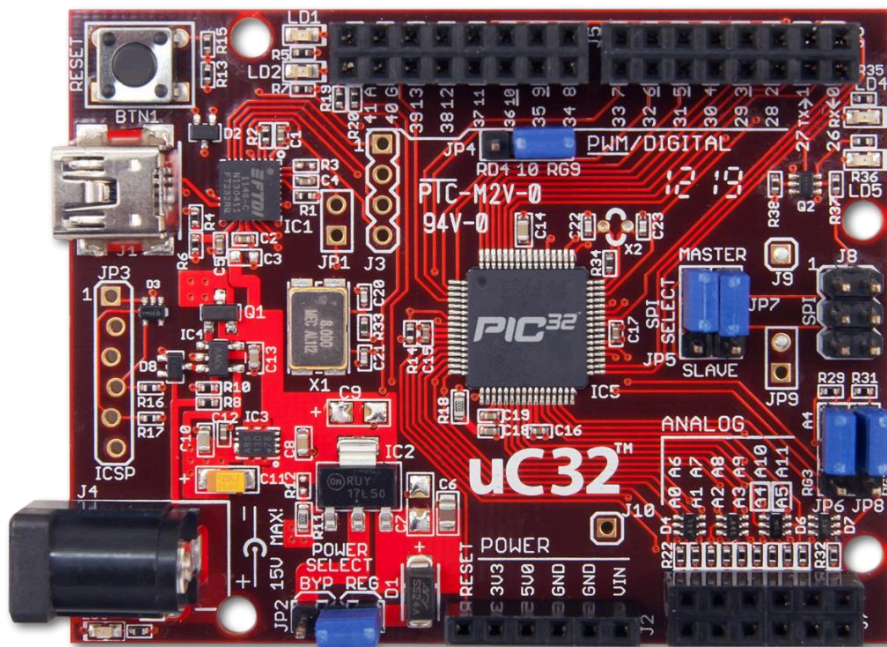
Semester:

Fall 2023



Objectives

1. Introduction of uC32 ChipKIT board, its components, and its features.
2. Introducing the Arduino IDE Software for programming microcontrollers like the uC32.
3. Hands-on experience in getting started with the uC32 ChipKIT board using the Arduino IDE.
4. Configuring the board, writing, and uploading code, and understanding of the output shown by the device or by the software.



The uC32 (also known as chipKIT uC32) is a microcontroller board based on the Microchip **PIC32MX340F512H** microcontroller. It's designed to be compatible with the **Arduino** while offering more processing power and additional features. The uC32 has various pins and ports that can be used for various purposes.

Features

Here's an explanation of the uC32 chipKIT pins and ports:



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I. Digital I/O Pins

The uC32 has a total of 42 digital I/O pins labeled from D0 to D41. These pins can be configured as either input or output and used for general-purpose digital input and output operations.

II. Analog Pins



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The uC32 features 12 analog input pins labeled A0 to A11 (pin 14-25). These pins can be used to read analog voltages from sensors and other analog devices. The PIC32 microcontroller on the uC32 has a built-in ADC (Analog-to-Digital Converter) for this purpose.

III. UART (Serial) Ports

The uC32 has two UART (Universal Asynchronous Receiver/Transmitter) ports for serial communication. These ports include:

- Serial (UART1): It uses pins 0 (RX) and 1 (TX).
- Serial1 (UART2): It uses pins 39 (RX) and 40 (TX).

IV. SPI Ports

The uC32 has SPI (Serial Peripheral Interface) ports that allow synchronous communication with other devices like sensors and displays. The SPI ports include SPI (pins 10-13).

V. I2C Ports

The PIC32 microcontroller shares pin A4, A5 with the two I2C signals SDA and SCL which can be used for connecting to I2C-compatible devices such as sensors and displays.

VI. PWM Pins

The uC32 provides PWM (Pulse Width Modulation) pins, which allow you to control the intensity of digital signals and are useful for tasks like motor control and LED dimming. These include pins 3, 5, 6, 9, and 10.

VII. External Interrupt Pins

The uC32 offers pins that can be configured as external interrupt inputs. These pins are labeled Pin 38 (INT0), Pin 2 (INT1), Pin 7 (INT2), Pin 8 (INT3), Pin 35 (INT4) and are typically used to trigger interruptions based on external events.

VIII. Power Pins

The uC32 has power pins for providing power to external components. These include V_{cc} (3.3V and 5V), GND (ground), and V_{in} (voltage input).

IX. USB Port

The uC32 has a USB port (USB-UART bridge) for serial communication and programming. It can be used to upload sketches and communicate with a computer.

These pins and ports provide a wide range of options for interfacing with sensors, displays, actuators, and other hardware components, making the uC32 chipKIT board suitable for a variety of embedded and IoT projects.

Getting started with the uC32 chipKIT board by Digilent involves a few key steps, including



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setting up the hardware, installing the necessary software tools, and writing and uploading your first program. Here's a step-by-step guide to help you get started:

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Hardware Setup

- **Unbox and Inspect**

Carefully unbox your uC32 chipKIT board and inspect it to ensure that all components are included and there is no physical damage.

- **Power Supply**

The uC32 board can be powered through a USB connection or an external power source (6-12V DC). If you plan to use external power, make sure you have the appropriate power adapter or battery pack.

- **Connect USB**

Connect the uC32 board to your computer using a USB cable. This cable will provide both power and a communication link between the board and your computer.

- **Install Drivers (if needed)**

Depending on your operating system, you may need to install drivers for the uC32 board. Drivers are provided on your google Classroom.

Software Setup

- **Arduino IDE**

Install the Arduino IDE from the official website: <https://www.arduino.cc/en/software>.

- **Install the Digilent Core**

To program the uC32 board using the Arduino IDE, you need to add the Digilent core. Here's how to do it:

- Open the Arduino IDE.

- Go to **File > Preferences**

- In the **Additional Boards Manager URLs** field, add the following URL:

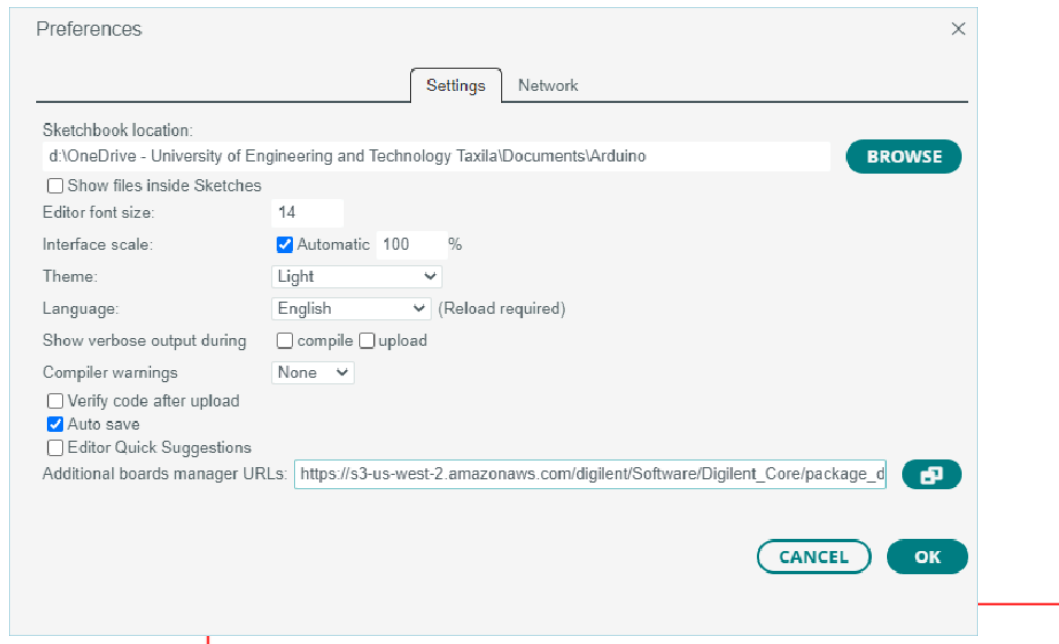
https://s3-us-west-2.amazonaws.com/digilent/Software/Digilent_Core/package_digilent_index.json



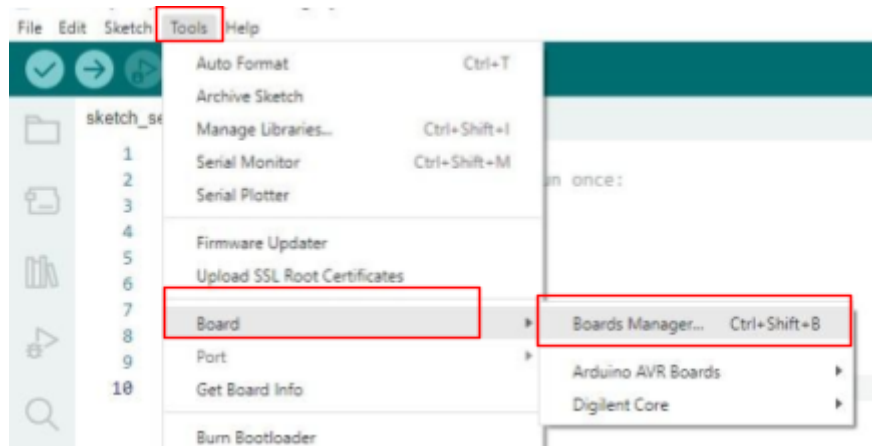
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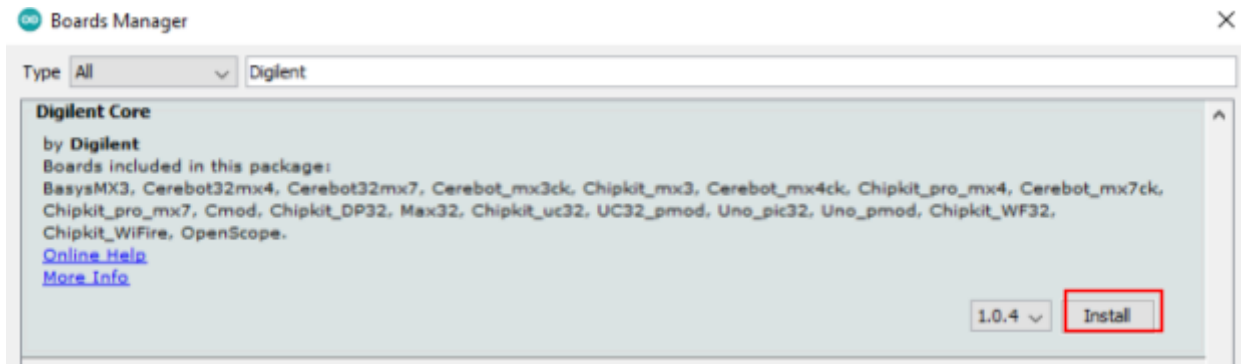
- Click **OK** to close the Preferences dialogue.
- Go to **Tools > Board > Boards Manager**





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- In the Boards Manager, search for **Digilent Core** and click the **Install** button next to the package.

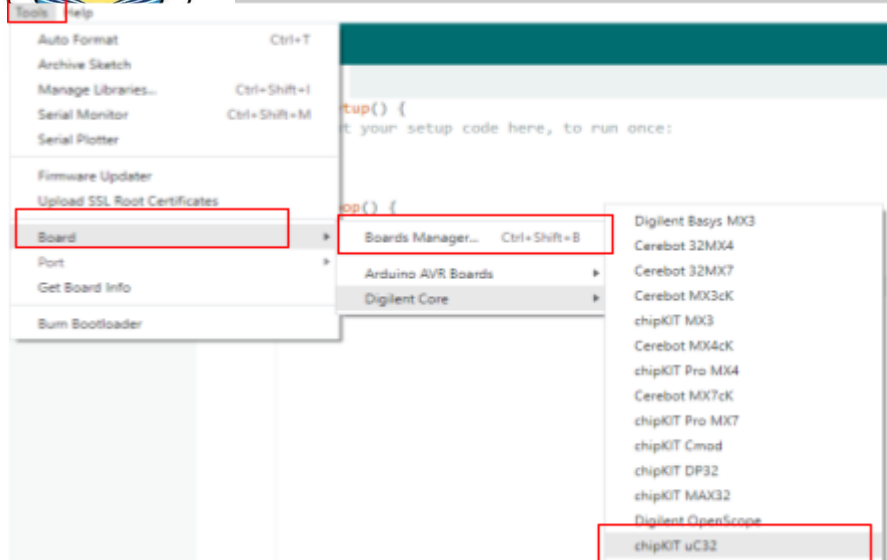


Write and Upload Your First Program

- Select the uC32 Board

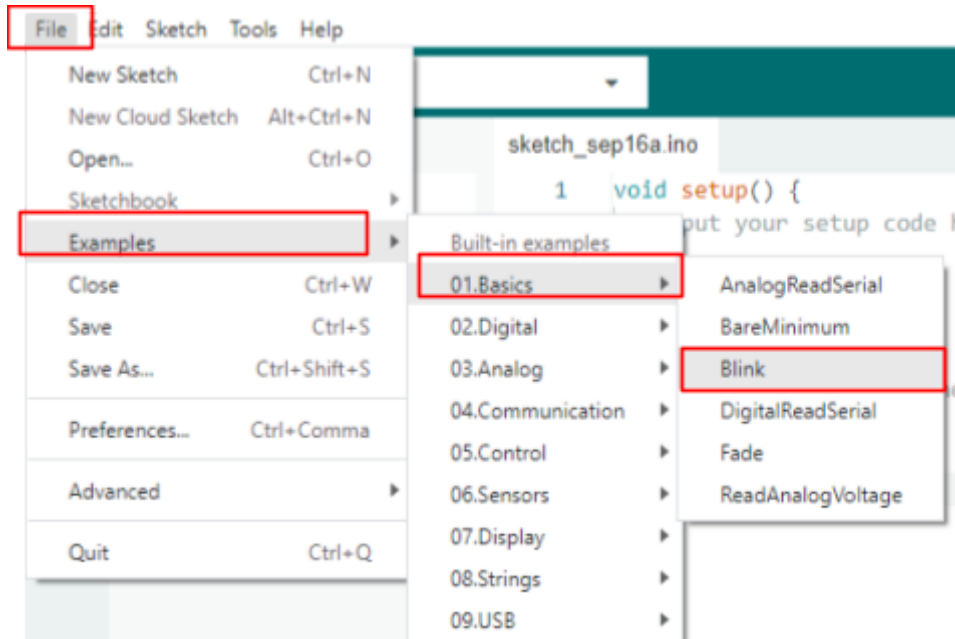
Now that you've installed the Digilent core, you can select the Digilent uC32 board from the Arduino IDE:

- Go to **Tools > Board** and you should see a list of Digilent boards. Select **chipKIT uC32**.





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Run Blink sketch by going to **File > Examples > Basics > Blink**.

▪ C Program for LED Blink

// the setup function runs once when you press reset or power the board

```
void setup() {
```

```
    // initialize digital pin LED_BUILTIN as an output.
```

```
    pinMode(13, OUTPUT);
```

```
}
```

// the loop function runs over and over again forever void loop() {

```
digitalWrite(13, HIGH); // turn the LED on (HIGH is the voltage level)
```

```
delay(1000); // wait for a second
```

```
digitalWrite(13, LOW); // turn the LED off by making the voltage LOW
```

```
delay(1000); // wait for a second
```



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}

- **Upload Your Program**

Click the **Upload** button (the right arrow icon) in the Arduino IDE to compile and upload your sketch to the uC32 board.

- **Test Your Program**

Once the program is uploaded, your uC32 board should execute it. Verify that led LD4 on uC32 board Blinking.

TASKS

1. Write down the PC/Laptop port name to which ChipKIT uC32 Board is connected.
2. Verify and upload Blink sketch on ChipKIT uC32 and write down the output.
3. Modify Blink sketch to blink led LD5 on the ChipKIT uC32 Board.
4. See the impact of Increasing/decreasing the delay on the Blinking process.

Lab Performance Evaluation:

Read and practice the manual. Performance is evaluated at the end of lab based on successfully running and understanding of examples, tasks and viva using defined rubrics.

Lab Report Evaluation:

Submit the Lab Report by writing all the examples and tasks which must include the following:

1. Brief description about how its work (3-5 lines)
2. The code
3. The results (Screenshot)

Important note: Lab Report must be according to the Lab Report Format available on Google Classroom and must be submitted on time. Two similar reports will get zero marks. So, don't try to copy your fellow reports. Lab report must be submitted in group of three maximum.



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